

SPONDIN AGCM, AB, Canada

WMO: 714130

Lat: 51.816N

Lon: 111.683W

Elev: 2638

StdP: 13.35

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.5	-20.1	-31.8	1.0	-25.3	-26.0	1.4	-19.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	86.2	61.4	82.4	60.9	78.8	59.4	65.0	77.3	63.2	76.0	61.6	74.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	88.3	69.8	58.8	81.6	68.1	56.9	76.1	65.7	31.4	77.3	30.0	75.5	28.8	74.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-32.4	92.0	4.2	4.2	-35.4	95.0	-37.9	97.5	-40.2	99.8	-43.3	102.8		
(5)			-33.1	69.2	4.2	2.1	-36.1	70.7	-38.6	72.0	-41.0	73.2	-44.0	74.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	12.9	9.9	23.2	37.0	49.9	57.7	62.9	61.4	52.0	38.5	23.4	10.6		
	DBStd	22.12	14.96	14.71	14.39	9.46	7.69	5.31	5.01	5.62	8.23	8.86	13.11	14.69		
	HDD50	6067	1150	1122	832	397	99	6	0	2	72	366	800	1221		
	HDD65	10378	1615	1542	1297	841	469	227	99	138	395	820	1249	1686		
	CDD50	1237	0	0	0	6	96	235	401	356	132	11	0	0		
	CDD65	73	0	0	0	0	1	6	35	27	4	0	0	0		
	CDH74	2100	0	0	0	7	106	196	829	702	258	2	0	0		
	CDH80	597	0	0	0	0	13	34	259	212	79	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	14.0	0.6	0.4	0.6	0.9	1.6	2.9	2.3	1.9	1.1	0.6	0.6	0.5		
	PrecMax	18.3	1.7	0.9	1.4	2.0	3.3	6.0	5.8	3.4	2.9	1.3	2.1	0.7		
(15) Precipitation	PrecMin	10.4	0.1	0.0	0.2	0.1	0.2	0.6	0.9	0.4	0.1	0.2	0.0	0.1		
	PrecStd	2.6	0.4	0.2	0.3	0.5	0.9	1.3	1.1	1.0	0.7	0.3	0.5	0.2		
	DB	46.0	44.3	58.2	71.1	81.6	84.4	91.5	90.9	87.1	71.5	58.6	44.8			
	MCWB	37.8	38.1	45.3	49.2	56.3	60.4	63.5	60.5	59.5	52.3	45.5	37.9			
	2% DB	40.5	37.7	52.1	65.6	76.9	79.2	86.4	85.4	81.8	64.7	50.4	39.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	35.6	33.3	41.5	47.9	54.2	59.3	63.0	61.0	58.3	49.8	41.3	34.1			
	5% DB	37.3	34.7	46.3	59.7	72.7	75.5	82.4	81.4	75.5	59.8	44.2	35.1			
	MCWB	33.3	31.7	38.6	44.1	53.1	58.8	62.2	60.5	56.2	46.9	37.3	30.9			
	10% DB	33.8	31.3	40.7	54.3	67.7	71.9	78.3	77.7	69.6	54.1	39.6	30.4			
	MCWB	30.8	29.2	35.4	41.5	50.7	57.3	61.0	60.0	53.6	44.0	34.6	27.6			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	38.7	37.5	46.6	51.9	58.1	64.9	69.6	66.6	62.0	54.8	46.7	38.9			
	MCDB	44.8	43.5	58.1	69.1	76.2	76.6	83.6	78.7	81.0	67.6	58.6	44.2			
	2% WB	35.7	33.9	42.1	47.8	56.0	62.2	65.9	64.5	59.2	51.0	41.7	34.5			
	MCDB	40.0	37.1	50.6	63.2	72.9	73.8	77.8	77.1	77.0	61.4	49.6	39.1			
	5% WB	33.7	31.8	38.7	44.9	54.1	60.3	63.8	62.6	57.0	47.7	38.2	31.3			
(30) Mean Daily Temperature Range	MCDB	37.0	34.6	45.8	57.5	69.5	72.0	76.2	76.1	72.3	58.5	43.1	34.9			
	10% WB	30.8	29.1	35.4	42.4	52.0	58.5	62.4	60.8	54.7	44.8	35.3	27.6			
	MCDB	33.7	31.4	40.2	52.9	65.3	69.0	75.0	73.8	69.1	53.8	38.9	30.4			
	5% DB	19.4	21.1	25.8	34.9	35.1	30.6	34.9	35.4	38.3	32.4	26.1	22.4			
	MCWBR	15.4	17.4	16.2	18.2	15.7	13.7	14.6	14.4	17.4	18.3	17.3	17.8			
(36) Clear-Sky Solar Irradiance	5% WB	20.1	20.6	24.9	31.1	29.9	26.0	29.0	30.1	35.1	30.7	24.3	22.5			
	MCWBR	16.5	17.4	16.3	17.1	14.2	13.0	14.1	13.9	16.7	17.6	16.8	18.3			
	taub	0.269	0.311	0.286	0.336	0.363	0.368	0.358	0.373	0.321	0.284	0.256	0.254			
	taud	2.306	2.217	2.462	2.342	2.313	2.345	2.410	2.338	2.460	2.540	2.483	2.322			
	Ebn at Noon	237	253	293	286	282	281	282	271	276	269	248	224			
(42) All-Sky Solar Radiation	Edh at Noon	20	29	28	35	38	37	35	35	28	21	16	16			
	RadAvg	361	696	1138	1525	1844	1919	2048	1666	1186	715	375	273			
(45)	RadStd	32	58	87	129	123	121	93	125	114	90	33	26			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page